

Mathematics for Computer Science

COMP1021 - Durham University

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Part I

Calculus

1 Introduction to Calculus

1.1 Sets

Definition 1.1.1: A set is an unordered collection of distinct elements denoted as $A = \{x | \text{condition}\}$.

Some common set operations include:

- **Definition 1.1.2:** The union of two sets, $A \cup B$, includes all the elements of both sets.
- **Definition 1.1.3:** The intersection of two sets, $A \cap B$, includes any elements which are present in both sets.
- **Definition 1.1.4:** If A and B are finite sets, then the principle of inclusion-exclusion states that

$$A \cup B = A + B - (A \cap B) \quad (1)$$

- **Definition 1.1.5:** The set A is a subset of B , $A \subseteq B$, if every element of A is an element of B . A is a proper subset of B , $A \subset B$, if A is a subset of B and there is at least one element of B not in A .
- **Definition 1.1.6:** The powerset of a set A is the set of all subsets of A .

$$\mathcal{P}(A) = \{A' | A' \subseteq A\} \quad (2)$$

- **Definition 1.1.7:** The cartesian product of two sets, $A \times B$, is the set of all possible ordered pairs with the first element from A and the second from B .

Definition 1.1.8: For a finite set, the cardinality is the number of distinct elements that it contains.

1.2 Functions

For a function $f : A \rightarrow B$,

- **Definition 1.2.1:** A is known as the domain of f .
- **Definition 1.2.2:** B is known as the codomain of f .
- **Definition 1.2.3:** If $f(a) = b$, then b is the image of a .
- **Definition 1.2.4:** The pre-image of b is the set $\{a | f(a) = b\}$.
- **Definition 1.2.5:** f forms an injection if every element of A maps to a unique element in B , even if some elements of B don't have a mapping.
- **Definition 1.2.6:** f forms a surjection if every element of B maps to at least one element of A .
- **Definition 1.2.7:** f forms a bijection if it is both injective and surjective.

Definition 1.2.8: If for an unbounded set A , there is a bijective function $f : A \rightarrow \mathbb{N}$, then we say A is countably infinite, otherwise A is known as uncountably infinite.

1.3 Hyperbolic Functions

Definition 1.3.1: Given the unit hyperbola $x^2 - y^2 = 1$, the hyperbolic functions are defined by the coordinates of a point at which area $\frac{1}{2}\alpha$ has been swept out.

$$\begin{aligned} \sinh(\alpha) &= \frac{e^\alpha - e^{-\alpha}}{2} & \cosh(\alpha) &= \frac{e^\alpha + e^{-\alpha}}{2} & \tanh(\alpha) &= \frac{e^\alpha - e^{-\alpha}}{e^\alpha + e^{-\alpha}} & \sigma(\alpha) &= \frac{1}{1 + e^{-\alpha}} \\ \operatorname{sech}(\alpha) &= \frac{2}{e^\alpha + e^{-\alpha}} & \operatorname{cosech}(\alpha) &= \frac{2}{e^\alpha - e^{-\alpha}} & \coth(\alpha) &= \frac{e^\alpha + e^{-\alpha}}{e^\alpha - e^{-\alpha}} \end{aligned}$$

These functions can then be differentiated as follows:

$$\begin{aligned} \frac{d}{dx} [\sinh(\alpha)] &= \cosh(\alpha) & \frac{d}{dx} [\cosh(\alpha)] &= \sinh(\alpha) \\ \frac{d}{dx} [\tanh(\alpha)] &= \operatorname{sech}^2(\alpha) & \frac{d}{dx} [\sigma(\alpha)] &= \sigma(\alpha)(1 - \sigma(\alpha)) \end{aligned}$$

2 Multivariate Functions

2.1 Multivariate Derivatives

Definition 2.1.1: For a function $f(x, y)$, if we take $f = f_0$ to be fixed but unknown, then we can differentiate f as a function of x at $x = x_0$, giving us the partial derivative of f with respect to x which gives the gradient when moving in the x -direction from any point.

$$\frac{\delta f}{\delta x} [f(x, y)] = \lim_{x \rightarrow x_0} \left(\frac{f(x, y_0) - f(x_0, y_0)}{x - x_0} \right) = f_x \quad (3)$$

Definition 2.1.2: The vector of partial derivatives, denoted as ∇f , contains all the partial derivatives of a function. This forms a vector pointing in the direction of the greatest rate of increase of f , where its magnitude is the rate of increase in that direction. For example, for a function $f(x, y)$,

$$\nabla f = \begin{pmatrix} f_x \\ f_y \end{pmatrix} \quad (4)$$

Definition 2.1.3: The directional derivative of the function $f(x, y)$ gives the gradient at any point whilst looking in the direction given by the unit vector $\mathbf{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix}$.

$$\nabla_{\mathbf{v}} f = \lim_{h \rightarrow 0} \frac{f(x_0 + h\mathbf{v}) - f(x_0)}{h} \quad (5)$$

If the slope is approximately flat at a small scale, then the slope can be determined by

$$\nabla_{\mathbf{v}} f = \mathbf{v} \cdot \nabla f \quad (6)$$

From this, we get that, for a function $f(\mathbf{x}_0)$ and a direction $\mathbf{v}$,

$$f(\mathbf{x}_0 + h\mathbf{v}) \approx f(\mathbf{x}_0) + h\mathbf{v} \cdot \nabla f(\mathbf{x}_0) \quad (7)$$

2.2 Multivariate Extrema

Definition 2.2.1: A function that is continuous is in class C^0 , and a function that is continuous and has a continuous derivative is in class C^1 . A function f such that f' , f'' , ..., $f^{(k)}$ exist and are continuous is in class C^k .

Definition 2.2.2: A function f such that derivatives of all orders exist and are continuous is in class C^∞ is also smooth.

Definition 2.2.3: All the second-order partial derivatives of a function $f(x, y)$ can be gathered into a Hessian matrix.

$$\mathbf{H}_f = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} \quad (8)$$

This can be used to determine the form of bivariate extrema. Suppose $f(x, y) \in C^2$ and $f_x(x_0, y_0) = 0 = f_y(x_0, y_0)$ then,

- (x_0, y_0) is a local maximum if $\det(\mathbf{H}_f) > 0$ and $f_{xx} < 0$ at (x_0, y_0) .
- (x_0, y_0) is a local minimum if $\det(\mathbf{H}_f) > 0$ and $f_{xx} > 0$ at (x_0, y_0) .
- (x_0, y_0) is a saddle point if $\det(\mathbf{H}_f) < 0$.
- If $\det(\mathbf{H}_f) = 0$, then the test is inconclusive.

At a stationary point x_0 , the quadratic approximation of a function $f(x)$ is

$$f(x_0 + \mathbf{v}) \approx f(x_0) + \frac{1}{2} \mathbf{v}^T \mathbf{H}_f(x_0) \mathbf{v} \quad (9)$$

But as $\mathbf{H}_f(x_0)$ is real and symmetric, it has orthogonal eigenvectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, so we can write

$$\mathbf{v} = c_1 \mathbf{e}_1 + c_2 \mathbf{e}_2 + \dots + c_n \mathbf{e}_n \quad (10)$$

When substituted back into the original equation, this gives that

$$f(x_0 + \mathbf{v}) \approx f(x_0) + \frac{1}{2} (\lambda_1(c_1)^2 + \lambda_2(c_2)^2 + \dots + \lambda_n(c_n)^2) \quad (11)$$

Given this, we can use the eigenvalues to determine the nature of the stationary point x_0 ,

- If all eigenvalues are positive, then the stationary point is a local minimum.
- If all eigenvalues are negative, then the stationary point is a local maximum.
- If there is a mixture of positive and negative eigenvalues, then the stationary point is a saddle point.

2.3 Vector-Valued Functions

Definition 2.3.1: A vector-valued function takes an n -dimensional vector as input, and outputs an m -dimensional vector, e.g., $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

Definition 2.3.2: The Jacobian matrix of a vector-valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be formed by setting $\mathbf{J}_{i,j} = \frac{\delta f_i}{\delta x_j}$ to form,

$$\mathbf{J}_f = \begin{pmatrix} \frac{\delta f_1}{\delta x_1} & \dots & \frac{\delta f_1}{\delta x_n} \\ \vdots & \ddots & \vdots \\ \frac{\delta f_m}{\delta x_1} & \dots & \frac{\delta f_m}{\delta x_n} \end{pmatrix} \quad (12)$$

For a multivariate function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the Jacobian matrix can be used to linearly approximate f at x ,

$$f(x + h\mathbf{v}) \approx f(x) + \mathbf{J}_f(x)(h\mathbf{v}) \quad (13)$$

3 Lagrange Multipliers

3.1 Constrained Optimisation

Definition 3.1.1: The goal of a constrained optimisation problem is to either minimise or maximise a multivariate function $f(\mathbf{x})$ where $x_1, x_2, \dots, x_n$ are not independent subject to some constraint $g(\mathbf{x})$.

Definition 3.1.2: The Lagrange multipliers method is used to find stationary points of an objective function $f(\mathbf{x}) \in C^1$ subject to the constraint $g(\mathbf{x}) = 0$.

1. Define a new objective function called the Lagrangian,

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda g(\mathbf{x}) \quad (14)$$

2. Find the stationary points of $\mathcal{L}$ with respect to both $\mathbf{x}$ and λ , i.e., solve the system of equations $\nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x})$ and $g(\mathbf{x}) = 0$ for the unknowns $\mathbf{x}$ and λ .
3. Then, the constrained extrema are found among the solutions to these equations. Identify their nature.

4 Series

4.1 Introduction to Series

Definition 4.1.1: A sequence is an ordered, countably infinite, set of numbers.

Definition 4.1.2: A series is the sum of the members of a sequence.

Theorem 4.1.1: If the series $\sum_{n=0}^{\infty} a_n$ and the series $\sum_{n=0}^{\infty} b_n$ are convergent, and have respective sums α and β , then

$$\sum_{n=0}^{\infty} (a_n + b_n) = \alpha + \beta \quad (15)$$

and

$$\sum_{n=0}^{\infty} (a_n - b_n) = \alpha - \beta \quad (16)$$

Proof: $\sum_{n=0}^{\infty} a_n$ converges to α and $\sum_{n=0}^{\infty} b_n$ converges to β . Given an arbitrary $\epsilon > 0$, let $\epsilon_a = \frac{1}{2}\epsilon = \epsilon_b$. Since $\sum_{n=0}^{\infty} a_n$ converges, $\exists N_a$ such that,

$$\left| \sum_{r=0}^n a_r - \alpha \right| < \epsilon_a, \quad \forall n > N_a \quad (17)$$

Similarly, since $\sum_{n=0}^{\infty} b_n$ converges, $\exists N_b$ such that,

$$\left| \sum_{r=0}^n b_r - \beta \right| < \epsilon_b, \quad \forall n > N_b \quad (18)$$

Taking $N = \max\{N_a, N_b\}$, then $\forall n > N$,

$$\begin{aligned} \left| \sum_{r=0}^n (a_r + b_r) - (\alpha + \beta) \right| &= \left| \left(\sum_{r=0}^n a_r - \alpha \right) + \left(\sum_{r=0}^n b_r - \beta \right) \right| \\ &\leq \left| \sum_{r=0}^n a_r - \alpha \right| + \left| \sum_{r=0}^n b_r - \beta \right| \leq \epsilon_a + \epsilon_b = \epsilon \end{aligned} \quad (19)$$

i.e, $\sum_{n=0}^{\infty} (a_n + b_n)$ converges to $(\alpha + \beta)$ using triangle inequality $|a + b| \leq |a| + |b|$.

□

Theorem 4.1.2: If $\lim_{n \rightarrow \infty} a_n \neq 0$ then the series $\sum_{n=0}^{\infty} a_n$ diverges.

Proof: Suppose $\sum_{n=0}^{\infty} a_n$ converges for some S , Then, $\lim_{n \rightarrow \infty} S_n = S$, which means that given arbitrary $\epsilon > 0$, there exists some N_ϵ such that, for all $n > N_\epsilon$,

$$|S_n - S| < \epsilon \quad |S_n - S| < 2\epsilon \quad (20)$$

Combining these,

$$2\epsilon > |S_n - S| + |S_{n+1} - S| = |S - S_n| + |S_{n+1} - S| \geq |S_{n+1} - S_n| = |a_{n+1}| \quad (21)$$

But ϵ was arbitrary, so we have shown that, for any $\epsilon' > 0$ there exists an N_ϵ such that $|a_n| < \epsilon'$ for all $n > N_\epsilon$.

□

As the number of elements in a series approaches infinity...

- **Definition 4.1.3:** If $\lim_{n \rightarrow \infty} a_n \neq 0$, then the series diverges.
- **Definition 4.1.4:** If $\lim_{n \rightarrow \infty} a_n = 0$, then the series converges.
- **Definition 4.1.5:** If $\lim_{n \rightarrow \infty} |a_n| = 0$, then the series absolutely converges.
- **Definition 4.1.6:** If $\lim_{n \rightarrow \infty} a_n = S$, then the series is convergent to S .

Theorem 4.1.3: The ratio test states that, if $\sum_{n=1}^{\infty} a_n$ is a series where $a_n \neq 0$ and $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ then

- If $L < 1$, the series absolutely converges.
- If $L > 1$, the series is divergent.
- If $L = 1$, then the test fails.

Proof: (a) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L < 1$, then $\exists r$ with $L < r < 1$ and N such that $\forall n > N$:

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < r \quad \rightarrow \quad |a_{n+1}| < r|a_n| \quad (22)$$

Then $|a_{n+2}| < r|a_{n+1}| < r^2|a_n|$ and $|a_{n+3}| < r|a_{n+2}| < r^3|a_n|$ etc. So the remainder is

$$R_{n+m} = \lim_{m \rightarrow \infty} \left(|a_n| r^{m+1} \frac{1}{1-r} \right) = 0 \quad (23)$$

Hence the series converges.

□

Definition 4.1.7: The n^{th} root test states that, for the series $\sum_{n=1}^{\infty} a_n$, $L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$,

- If $L < 1$, then the series absolutely converges.
- If $L > 1$, then the series is divergent.
- If $L = 1$, then the test fails.

Definition 4.1.8: The alternating series test states that the series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ converges if $a_n > 0$, $a_{n+1} \leq a_n \forall n$, and $\lim_{n \rightarrow \infty} a_n = 0$

Theorem 4.1.4: Every subsequence of a convergent sequence converges to the same limit as the original sequence.

Proof: Let $\{a_{n_k}\}$ be a subsequence of $\{a_n\}$. First note that $n_k \geq k$ and so $n_{k+1} \geq k+1$. Let $\lim_{n \rightarrow \infty} a_n = L$. Then, for any $\epsilon > 0$, there exists N_ϵ such that for all $n > N_\epsilon$ it holds that $|a_n - L| < \epsilon$. Let $k > N_\epsilon$, then $n_k > N_\epsilon$ and so it must be that $|a_{n_k} - L| < \epsilon$.

□

4.2 Power Series

Definition 4.2.1: A power series is a series involving a variable x and a constant x_0 of the form

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \dots \quad (24)$$

At any specific value of x , the power series becomes a normal series.

Definition 4.2.2: For an arbitrary power series $\sum_{n=0}^{\infty} a_n (x - x_0)^n$, we can use the ratio test (including the $(x - x_0)$ term). From this, we call $\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = r$ the radius of convergence. If $|x - x_0| < r$, we have absolute convergence and if $|x - x_0| > r$, we have divergence.

Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence $r > 0$. Then, within the interval $(-r, r)$:

- $f(x)$ is continuous.
- $f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$ (term-by-term differentiation).
- $\int_0^x f(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$ (term-by-term integration).
- Both series $f'(x)$ and $\int_0^x f(t) dt$ have the same radius of convergence as $f(x)$.

Definition 4.2.3: Suppose we start with some function $f(x)$ and we wish to determine a power series $\sum_{n=0}^{\infty} a_n x^n$ such that $f(x) = \sum_{n=0}^{\infty} a_n x^n$. Putting these values in, we see that if such a power series exists, then:

$$f(x) = f(0) + f'(0)x + f''(0)\frac{x^2}{2!} + f^{(3)}(0)\frac{x^3}{3!} + \dots = \sum_{n=0}^{\infty} f^{(n)}(0)\frac{x^n}{n!} \quad (25)$$

This is called the Maclaurin series for f . To use the series, we must have that $f \in C^\infty$ and the power series converges.

Definition 4.2.4: We can recenter the power series on x_0 by defining a new function $g(x) = f(x + x_0)$ then $g(0) = f(x_0)$ and $g'(x) = f'(x + x_0)$. Repeatedly differentiating, $g^{(n)}(x) = f^{(n)}(x + x_0)$ so we build the Maclaurin series for g , which gives us

$$f(x + x_0) = g(x) = \sum_{n=0}^{\infty} g^{(n)}(0)\frac{x^n}{n!} = \sum_{n=0}^{\infty} f^{(n)}(x_0)\frac{x^n}{n!} \quad (26)$$

or,

$$f(x) = g(x - x_0) = f(x_0) + f'(x_0)(x - x_0) + f''(x_0)\frac{(x - x_0)^2}{2!} + \dots \quad (27)$$

This form is called the Taylor series expansion of f .

4.3 Taylor's Theorem

Lemma 4.3.1: If a real-valued function f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists $\xi \in (a, b)$ such that $f'(\xi) = 0$.

Proof: Not proved here. □

Theorem 4.3.1: Suppose a function f is n times differentiable on $[a, x]$, then there exists some $\xi \in (a, x)$ such that

$$f(x) = f(a) + (x-a)f'(a) + \dots + \frac{(x-a)^{(n-1)}}{(n-1)!}f^{(n-1)}(a) + R^n(x) \quad (28)$$

Where the remainder term is equal to $R_n(x) = \frac{(x-a)^n}{n!}f^{(n)}(\xi)$.

Proof: Suppose a function f is n times differentiable on $[a, x]$. Let k be a constant defined so that

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots + \frac{(x-a)^{(n-1)}}{(n-1)!}f^{(n-1)}(a) + \frac{(x-a)^n}{n!}k \quad (29)$$

We can then define the function

$$F(y) = f(x) - f(y) - (x-y)f'(y) - \dots - \frac{(x-y)^{(n-1)}}{(n-1)!}f^{(n-1)}(y) - \frac{(x-y)^n}{n!}k \quad (30)$$

Then $F(x) = 0$ and also $F(a) = f(x) - f(x) = 0$ so we can apply Rolle's theorem to F on $[a, x]$, meaning there is some $\xi \in (a, x)$ such that $F'(\xi) = 0$. □

4.4 Fourier Series

Definition 4.4.1: For a function $f(x)$, $-\pi < x < \pi$, the Fourier series is

$$f(x) \approx \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \quad (31)$$

where

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx \quad (32)$$

Part II

Linear Algebra

5 Introduction to Linear Algebra

5.1 Mathematical Structures

Definition 5.1.1: Abstraction is the process of simplifying a complex problem via removing unnecessary details in order to apply more general problem solving techniques.

Definition 5.1.2: Groups consist of a set G and a binary operation $*$. They abide by the following axioms:

- **Definition 5.1.3:** Closure, when applying an operation between two elements of the group, the result will also be an element of the group.

$$\forall a, b \in G, a * b = c, c \in G \quad (33)$$

- **Definition 5.1.4:** Associativity, for any elements a, b, c in the group, the order in which the operation is performed does not matter, as long as the elements stay in their original order.

$$\forall a, b, c \in G, a * (b * c) = (a * b) * c \quad (34)$$

- **Definition 5.1.5:** Identity, the group contains an element such that when an operation is performed between an element in the group and the identity, the original element is returned.

$$\exists e \in G, \forall a \in G, a * e = e * a = a \quad (35)$$

- **Definition 5.1.6:** Inverse, for every element in the group, there exists an inverse element such that when an operation is performed between them, the identity is returned.

$$\forall a \in G, \exists b \in G, a * b = e \quad (36)$$

Definition 5.1.7: Fields contain a set F with two binary operations (addition and multiplication) and abide by the following axioms:

- Associativity (defined above).
- **Definition 5.1.8:** Commutativity - when applying an operation between two elements of the field, the order the elements appear in does not matter.

$$\forall a, b \in F, a + b = b + a \text{ and } a \times b = b \times a \quad (37)$$

- **Definition 5.1.9:** Identity, there exists two distinct elements, 0 and 1, in F such that,

$$\exists 0, 1 \in F, \forall a \in F, a + 0 = a \text{ and } a \times 1 = a \quad (38)$$

- **Definition 5.1.10:** Additive inverses, for every element in the field, there exists some other element, called the additive inverse, such that,

$$\forall a \in F, \exists b \in F, a + b = 0 \quad (39)$$

- **Definition 5.1.11:** Multiplicative inverses, for every element in the field, there exists some other element, called the multiplicative inverse, such that,

$$\forall a \in F, \exists b \in F, a \times b = 1 \quad (40)$$

- **Definition 5.1.12:** Distributivity of multiplication over addition. This axiom states that,

$$\forall a, b, c \in F, a \times (b + c) = (a \times b) + (a \times c) \quad (41)$$

Definition 5.1.13: A vector space is a mathematical structure with two sets, scalars F and vectors V , where V forms a commutative group under addition and F forms a field. They abide by the following axioms:

- Associativity (defined above).
- Identity (defined above).
- Distributivity of multiplication over addition (defined above).

5.2 Linear Combinations and Maps

Definition 5.2.1: Given vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ and scalars $a_1, a_2, \dots, a_n$, we can form a linear combination.

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n = \sum_{i=1}^n a_i\mathbf{v}_i \quad (42)$$

Definition 5.2.2: The set of all linear combinations of a set of vectors is called its span, denoted $\text{span}(V)$.

Definition 5.2.3: A set of vectors S is a spanning set for a vector space V if $\text{span}(S) = V$. We can equivalently say that S spans V .

Definition 5.2.4: A set of vectors S is linearly dependent if there exists a linear combination such that

$$\sum_{i=1}^n a_i\mathbf{s}_i = 0 \text{ and } \exists i, a_i \neq 0 \quad (43)$$

Otherwise, the set is described as linearly independent.

Definition 5.2.5: A subset S of the vector space V is a basis for V if S is a spanning set and is linearly independent.

Theorem 5.2.1: For a vector space V , if S is a basis for V , then it must also be a minimal spanning set.

Proof: Let S be a basis of V . Assume that S is not a minimal spanning set, then there must be some element $s \in S$ such that $\text{span}(S \setminus \{s\}) = V$. Then, s can be written as a linear combination of elements from $S \setminus \{s\}$. That is, we have that

$$1 \cdot \mathbf{s} - (a_1\mathbf{s}_1 + a_2\mathbf{s}_2 + \dots + a_n\mathbf{s}_n) = 0 \quad (44)$$

This means that S would be linearly dependent, contradicting the definition of S as a basis. Therefore, S must be a minimal spanning set. □

Definition 5.2.6: The dimension of a vector space V is the cardinality of its basis S .

Definition 5.2.7: If we have a function $f : V \rightarrow W$ where V and W are vector spaces over a field F , then f is a linear map if,

$$\exists \mathbf{u}, \mathbf{v} \in V, \exists a \in F, f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v}) \text{ and } f(a \cdot \mathbf{v}) = a \cdot f(\mathbf{v}) \quad (45)$$

These can also be called linear transformations or vector space homomorphisms.

Definition 5.2.8: If $V = W$ for a linear map f , then f is also a endomorphism.

6 Matrices

6.1 Matrix Operations

6.2 Systems of Equations and Gaussian Elimination

6.3 Geometry

7 Decomposition

Part III

References

8 Module Professors

The below is accurate as of the 2025/26 academic year and will not be updated in subsequent years.

- *Calculus* is taught by Doctor Junyan Hu and Doctor Sukanya Pandey.

Doctor Junyan Hu:

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- *Linear Algebra* is taught jointly by Professor Steven Bradley and Doctor Karl Southern.

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9 Cited Works